Title: VULNERABILITY ASSESSMENT REPORT
Subtitle: Assessment Target: scanme.nmap.org
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Submitted to: FUTURE INTERNS

Executive Summary

- This report presents the results of a vulnerability assessment conducted on a publicly accessible website. The objective was to identify potential security weaknesses through passive and non-intrusive techniques.
- The assessment revealed several medium-risk issues, primarily related to outdated services, exposed network ports, and unencrypted communication. While no critical vulnerabilities were identified, these findings increase the system's exposure to potential cyber threats.
- Addressing the identified issues will significantly improve the system's security posture, reduce attack surface, and enhance data protection.

Scope & Methodology

Scope

- Public-facing services only
- No authentication or login testing
- No exploitation or active attacks performed

Methodology

- The assessment followed ethical security auditing practices, focusing on passive reconnaissance and configuration analysis

Tools used

- Nmap (network scanning and service detection)
- OWASP ZAP (Passive Scan) - Identifying vulnerabilities without attacking
- Firefox DevTools (header and client-side inspection)

Technical Overview

The target system was scanned to identify:

- Open ports
- Running services
- Service versions
- Exposure level

Summary of Open ports

Port	Service	Description
22	SSH	Remote Access Service
80	HTTP	Web Server
9929	Nping Echo	Network testing service
31337	Unknown (tcpwrapped)	Restricted or custom service

Findings & Risk Analysis

1. Outdated SSH Service

- The system is running an outdated version of SSH (OpenSSH 6.6.1)

Outdated software may contain known vulnerabilities that attackers can exploit to gain unauthorized access.

- Risk Level: Medium

Recommendation:

- Upgrade SSH to the latest version
- Disable password authentication and use SSH keys
- Restrict access to trusted IP addresses

2. Outdated Web Server

- The web server is running an older version of Apache (2.4.7).

Older web server versions may be vulnerable to publicly known exploits, increasing the risk of website compromise.

- Risk Level: Medium

Recommendation:

- Update Apache to the latest stable version
- Apply regular security patches
- Consider deploying a Web Application Firewall

3. Unencrypted Communication (HTTP)

- The website is accessible over HTTP without enforcing HTTPS.

Data transmitted between users and the server can be intercepted or altered by attackers.

- Risk Level: Medium

Recommendation:

- Implement HTTPS using SSL/TLS certificates
- Redirect all HTTP traffic to HTTPS

4. Exposure of Non-standard Ports

- Ports 9929 and 31337 are open and publicly accessible.

Uncommon services increase the attack surface and may not be properly monitored or secured.

- Risk Level: Medium

Recommendation:

- Disable unnecessary services
- Restrict access using firewall rules
- Monitor unusual network activity

5. Information Disclosure

- The system reveals service and version information during scanning.

Attackers can use this information to identify and target known vulnerabilities.

- Risk Level: Low

Recommendation:

- Disable service version disclosure
- Configure servers to return minimal information

Risk Summary

Finding	Risk Level
Outdated SSH Service	Medium
Outdated Apache Server	Medium
Unencrypted HTTP	Medium
Open Non-Standard Ports	Medium
Information Disclosure	Low

Recommendations

Immediate Actions

- Enable HTTPS across the website
- Update outdated services (SSH, Apache)
- Restrict remote access (SSH)

Short-term Improvements

- Close or secure unnecessary open ports
- Implement firewall rules
- Reduce information exposure

Long-Term Strategy

- Perform regular vulnerability assessments
- Establish patch management processes
- Implement continuous monitoring and logging

Conclusion

- The assessment identified several security weaknesses that, while not critical, contribute to an increased attack surface. These include outdated software, exposed services, and lack of encryption.
- By implementing the recommended controls, the organization can significantly enhance its security posture, reduce risks, and ensure better protection of systems and data.